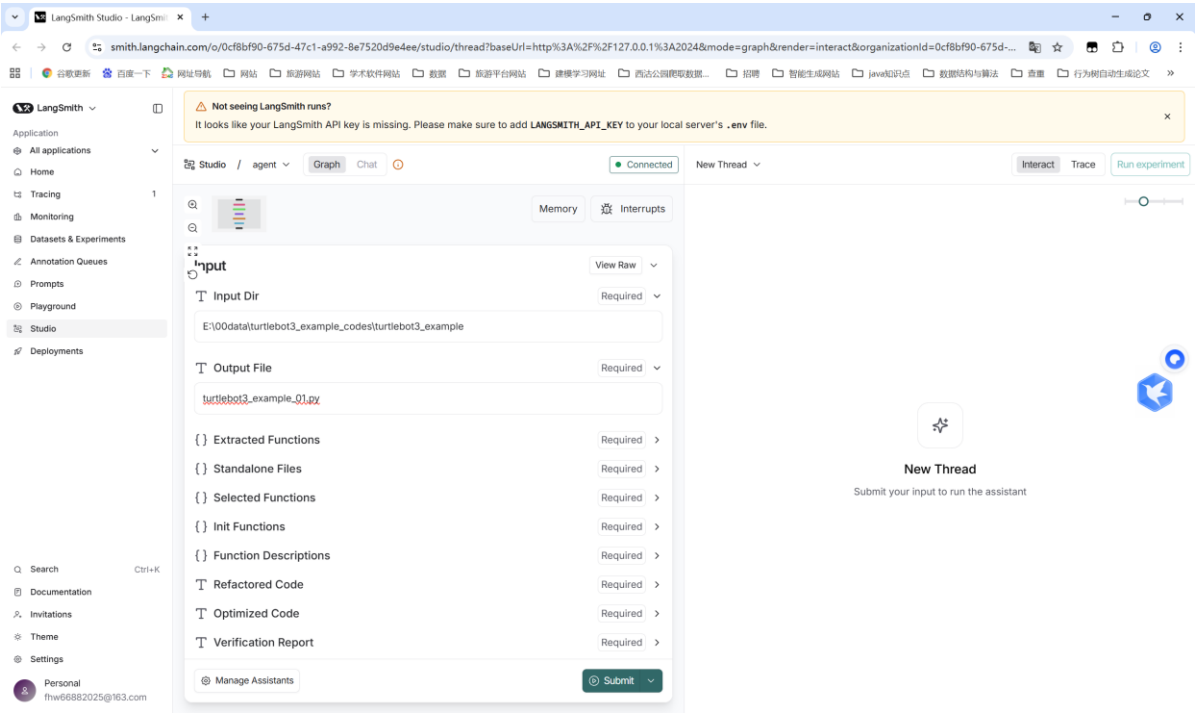
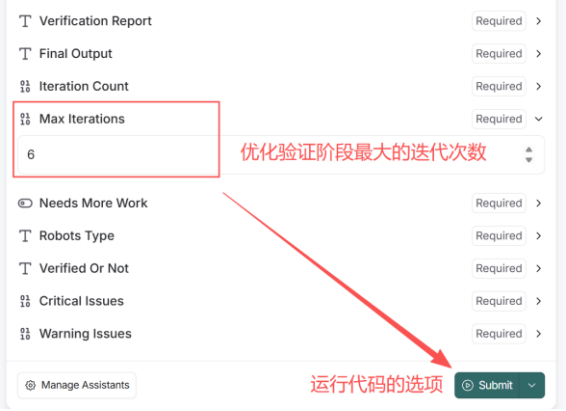
Appendix: End-to-End Execution Example of AutoRSP

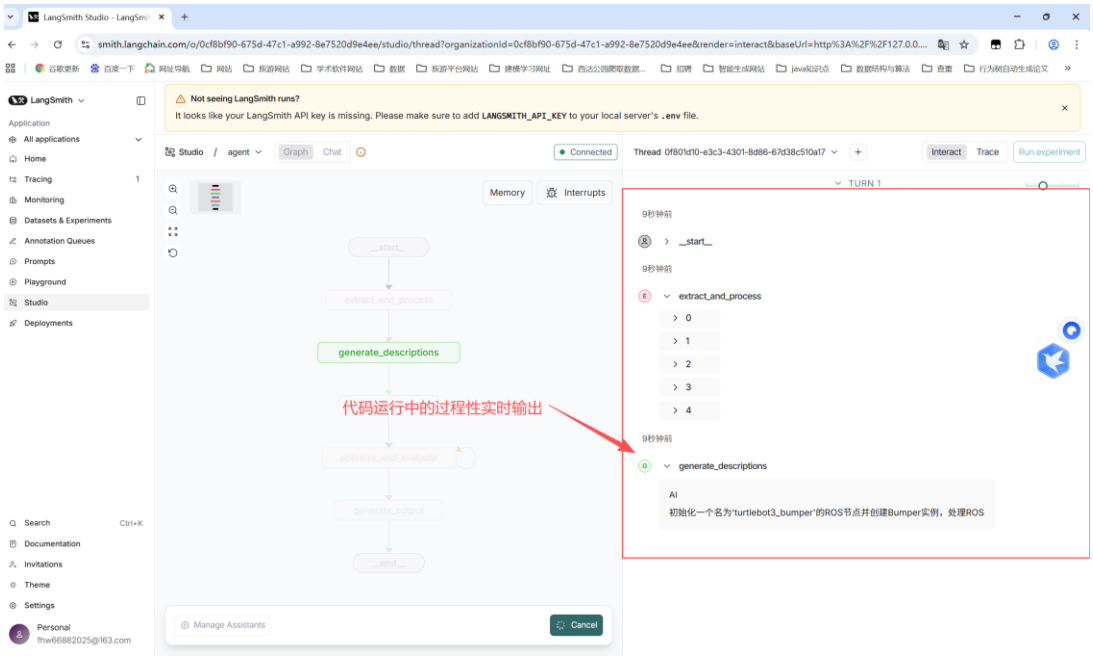
This appendix presents a **fully reproducible end-to-end execution example** of AutoRSP, illustrating the complete pipeline from raw robotic code input to a verified Robot Skill Primitive (RSP) class output. The example uses the *turtlebot3_example* dataset (consistent with the main paper) and includes runtime logs, workflow visualizations, and intermediate outputs to ensure full transparency and replicability for reviewers.

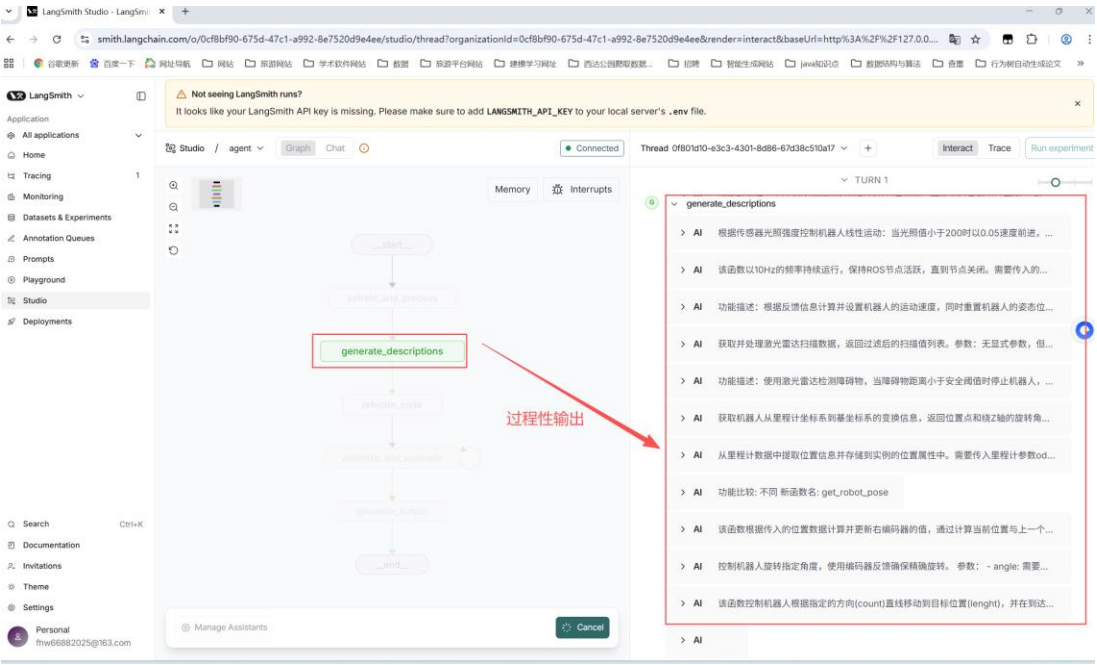
A. **The step-by-step output from the LangGraph web interface is captured in the following screenshot:**

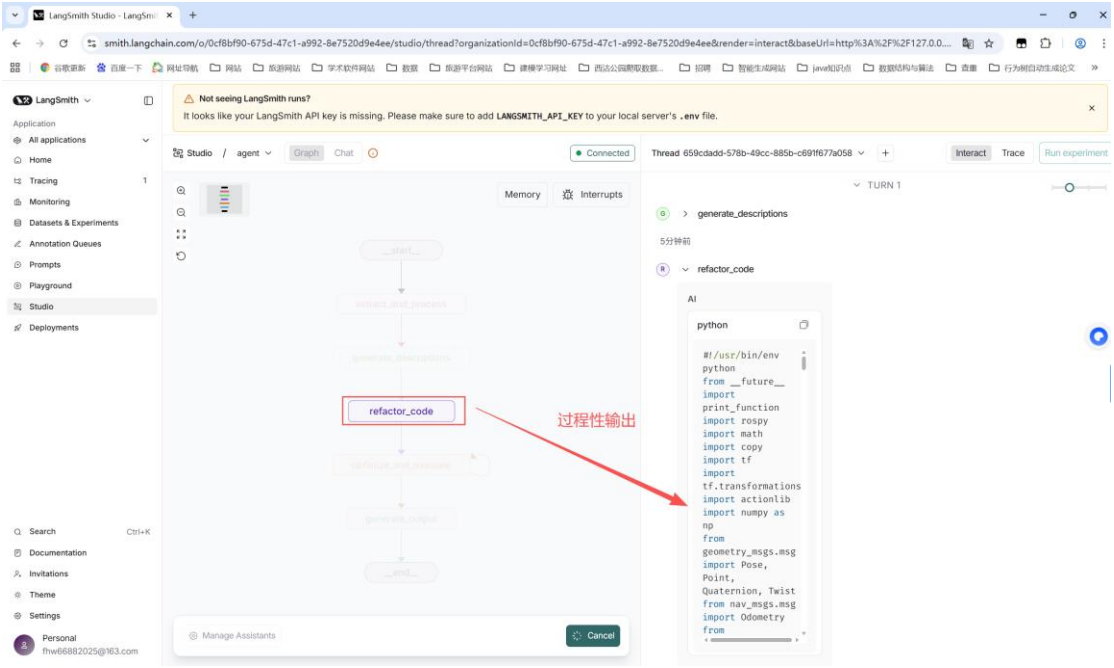


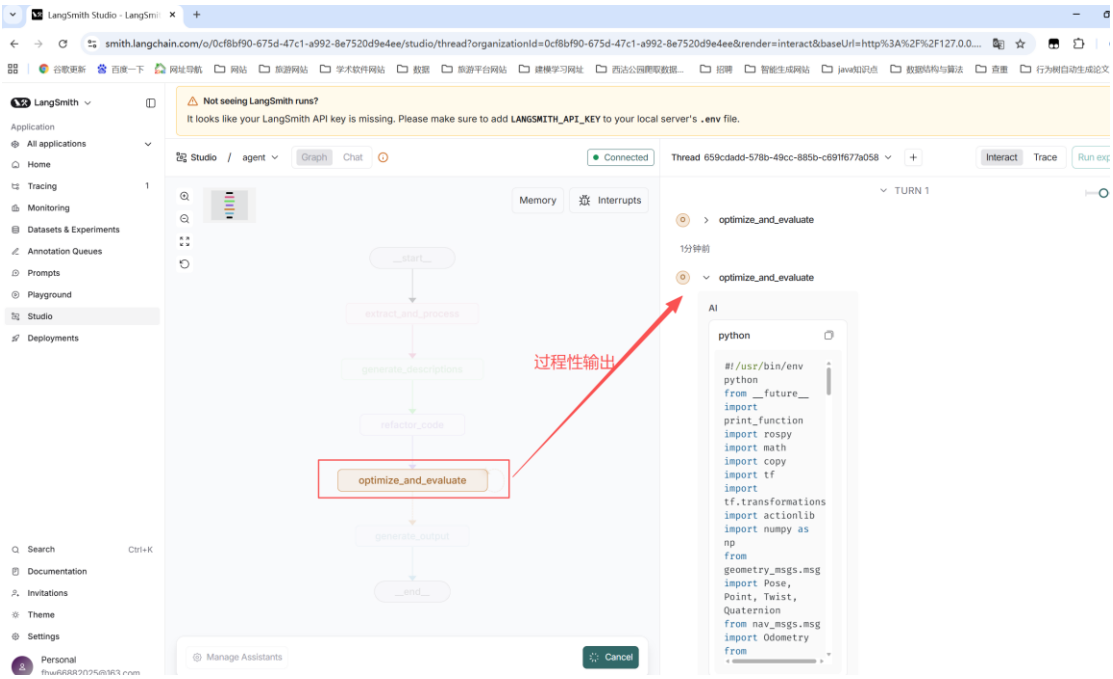
Input
View Raw
Input Dir
Required
源代码路径
E:\00data\turtlebot3_example_codes\turtlebot3_example
Output File
最终RSP的输出路径：项目同目录下
Required
turtlebot3_example_01.py
Extracted Functions
Standalone Files
Selected Functions
Init Functions
Function Descriptions
Refactored Code
Optimized Code
Verification Report
langgraph State在代码运行时保存的一些变量


Verification Report
Final Output
Iteration Count
Max Iterations
6
优化验证阶段最大的迭代次数
Needs More Work
Robots Type
Verified Or Not
Critical Issues
Warning Issues
Required
Manage Assistants
运行代码的选项
Submit


Not seeing LangSmith runs?
It looks like your LangSmith API key is missing. Please make sure to add LANGSMITH_API_KEY to your local server's .env file.
Memory
Interrupts
generate_descriptions
代码运行中的过程性实时输出
TURN 1
__start__
extract_and_process
generate_descriptions
初始化一个名为'turtlebot3_bumper'的ROS节点并创建Bumper实例，处理ROS
Manage Assistants
Cancel


Not seeing LangSmith runs?
It looks like your LangSmith API key is missing. Please make sure to add LANGSMITH_API_KEY to your local server's .env file.
Memory
Interrupts
generate_descriptions
过程性输出
TURN 1
Manage Assistants
Cancel


LangSmith Studio - LangSmi
smith.langchain.com/o/0cf8bf90-675d-47c1-a992-8e7520d9e4ee/studio/thread?organizationId=0cf8bf90-675d-47c1-a992-8e7520d9e4ee&render=interact&baseUrl=http%3A%2F%2F127.0.0....
LangSmith
Application
All applications
Home
Tracing
Monitoring
Datasets & Experiments
Annotation Queues
Prompts
Playground
Studio
Deployments
Search
Ctrl+K
Documentation
Invitations
Theme
Settings
Personal
fhw66882025@163.com
Not seeing LangSmith runs?
It looks like your LangSmith API key is missing. Please make sure to add LANGSMITH_API_KEY to your local server's .env file.
Studio / agent
Graph
Chat
Connected
Thread 659cdadd-578b-49cc-885b-c691f677a058
Interact
Trace
Run experiment
Memory
Interrupts
TURN 1
generate_descriptions
5分钟前
refactor_code
AI
python
#!/usr/bin/env
python
from __future__
import
print_function
import rospy
import math
import copy
import tf
import
tf.transformations
import actionlib
import numpy as
np
from
geometry_msgs.msg
import Pose,
Point,
Quaternion, Twist
from nav_msgs.msg
import Odometry
from
refactor_code
过程性输出
Manage Assistants
Cancel


LangSmith Studio - LangSmi
smith.langchain.com/o/0cf8bf90-675d-47c1-a992-8e7520d9e4ee/studio/thread?organizationId=0cf8bf90-675d-47c1-a992-8e7520d9e4ee&render=interact&baseUrl=http%3A%2F%2F127.0.0....
LangSmith
Application
All applications
Home
Tracing
Monitoring
Datasets & Experiments
Annotation Queues
Prompts
Playground
Studio
Deployments
Search
Ctrl+K
Documentation
Invitations
Theme
Settings
Personal
Not seeing LangSmith runs?
It looks like your LangSmith API key is missing. Please make sure to add LANGSMITH_API_KEY to your local server's .env file.
Studio / agent
Graph
Chat
Connected
Thread 659cdadd-578b-49cc-885b-c691f677a058
Interact
Trace
Memory
Interrupts
TURN 1
optimize_and_evaluate
1分钟前
optimize_and_evaluate
AI
python
#!/usr/bin/env
python
from __future__
import
print_function
import rospy
import math
import copy
import tf
import
tf.transformations
import actionlib
import numpy as
np
from
geometry_msgs.msg
import Pose,
Point, Twist,
Quaternion
from nav_msgs.msg
import Odometry
from
过程性输出
optimize_and_evaluate
Manage Assistants
Cancel

B.The runtime log output from the PyCharm Integrated Development Environment (IDE) terminal during code execution is presented below.

```
Project    graph_08.py
Terminal   Local
-67d38c510a17 thread_name=MainThread
2026-01-26T00:05:57.482941Z [info     ] Starting background run        [langgraph_api.worker] api_variant=local_dev assistant_id=fe096781-56
01-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 request_id=a4a74dd8-1a2e-4be9-bc92-994e3b081b49 resumable=False run_at
tempt=1 run_creation_ms=2 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf run_queue_ms=905 run_started_at=2026-01-26T00:05:57.482941+00:00 run_s
tream_start_ms=0 temporary=False thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=asyncio_0
2026-01-26T00:05:57.494666Z [info     ] 🔍 开始解析目录: E:\00data\turtlebot3_example_codes\turtlebot3_example [__src__agent__graph_08] api_
variant=local_dev assistant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=extract_and_p
rocess request_id=a4a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-430
1-8d86-67d38c510a17 thread_name=asyncio_0
2026-01-26T00:05:57.516669Z [info     ] ✅提取完成! 在 9 个文件中发现 35 个函数和 0 个独立文件 [__src__agent__graph_08] api_variant=local_dd
ev assistant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=extract_and_process request_
id=a4a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c51
0a17 thread_name=asyncio_0
2026-01-26T00:05:57.521669Z [info     ] 🧠 开始生成函数描述并处理重名函数...          [__src__agent__graph_08] api_variant=local_dev assist
ant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=generate_descriptions request_id=a4a7
4dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 th
read_name=asyncio_1
```

```
2026-01-26T00:06:29.137608Z [info     ]  ✓ 重命名函数: main -> createTurtlebotNode (来自 E:\00data\turtlebot3_example_codes\turtlebot3_exam
ple\turtlebot3_cliff.py) [__src__agent__graph_08] api_variant=local_dev assistant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent lan
ggraph_api_version=0.4.31 langgraph_node=generate_descriptions request_id=a4a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f
-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=asyncio_1
```

```
2026-01-26T00:09:05.157360Z [info     ]  ✓ 重命名函数: getkey -> checkAndShutdown (来自 E:\00data\turtlebot3_example_codes\turtlebot3_examp
le\turtlebot3_pointop_key.py) [__src__agent__graph_08] api_variant=local_dev assistant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agen
t langgraph_api_version=0.4.31 langgraph_node=generate_descriptions request_id=a4a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019
bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=asyncio_1
```

```
2026-01-26T00:26:49.618153Z [info     ]  发现 8 个__init__函数, 准备合并        [__src__agent__graph_08] api_variant=local_dev assistant_id
=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=generate_descriptions request_id=a4a74dd8-1
a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_n
ame=asyncio_1
```

```
2026-01-26T00:28:29.423573Z [info     ] ✅降重后唯一函数数量: 28                [__src__agent__graph_08] api_variant=local_dev assistant_idd
=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=generate_descriptions request_id=a4a74dd8-1
a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_n
ame=asyncio_1
2026-01-26T00:28:29.428832Z [info     ] 🔄 开始重构为统一类...                  [__src__agent__graph_08] api_variant=local_dev assistant_id=
fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=refactor_code request_id=a4a74dd8-1a2e-4be9-
bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=async
io_3
```

```
2026-01-26T00:31:28.810510Z [info     ] ✅重构成功! 生成统一类代码               [__src__agent__graph_08] api_variant=local_dev assistant___
id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=refactor_code request_id=a4a74dd8-1a2e-4b
e9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=as
yncio_3
2026-01-26T00:31:28.818430Z [info     ] 🔄 当前迭代次数: 1/6                  [__src__agent__graph_08] api_variant=local_dev assistant_id=fe
096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e
-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name
=asyncio_1
2026-01-26T00:31:28.819508Z [info     ] 🔬 开始代码验证...                    [__src__agent__graph_08] api_variant=local_dev assistant_id=fe
096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e
-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name
=asyncio_1
```

```
2026-01-26T00:32:25.437994Z [info     ] ✅验证完成!                          [__src__agent__graph_08] api_variant=local_dev assistant_id=fe099
6781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e-4
be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=a
syncio_1
2026-01-26T00:32:25.437994Z [info     ] 📊 分析验证报告，决定是否需要继续优化...         [__src__agent__graph_08] api_variant=local_dev assi
stant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4
a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17
thread_name=asyncio_1
2026-01-26T00:32:25.438993Z [info     ] 🔧 发现 5 个严重问题，需要继续优化            [__src__agent__graph_08] api_variant=local_dev assista
nt_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74
dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thr
ead_name=asyncio_1
2026-01-26T00:32:25.444502Z [info     ] 🔄 当前迭代次数: 2/6                  [__src__agent__graph_08] api_variant=local_dev assistant_id=fe
096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e
-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name
=asyncio_3
2026-01-26T00:32:25.445502Z [info     ] 🛠 检测到验证报告，进入优化模式...           [__src__agent__graph_08] api_variant=local_dev assista
n                                                                                                                                          n
```

```
2026-01-26T00:35:52.973653Z [info     ] ✅优化成功! 生成优化类代码                [__src__agent__graph_08] api_variant=local_dev assistant__
id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8
-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread
_name=asyncio_3
2026-01-26T00:35:52.974654Z [info     ] 🔬 开始代码验证...                    [__src__agent__graph_08] api_variant=local_dev assistant_id=fe
096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e
-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name
=asyncio_3
```

```
2026-01-26T00:36:38.932206Z [info     ] ✅验证完成!                          [__src__agent__graph_08] api_variant=local_dev assistant_id=fe099
6781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e-4
be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=a
syncio_3
2026-01-26T00:36:38.932206Z [info     ] 📊 分析验证报告，决定是否需要继续优化...         [__src__agent__graph_08] api_variant=local_dev assi
stant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4
a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17
thread_name=asyncio_3
2026-01-26T00:36:38.932206Z [info     ] 🔧 发现 1 个严重问题，需要继续优化            [__src__agent__graph_08] api_variant=local_dev assista
nt_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74
dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thr
ead_name=asyncio_3
2026-01-26T00:36:38.942407Z [info     ] 🔄 当前迭代次数: 3/6                  [__src__agent__graph_08] api_variant=local_dev assistant_id=fe
096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e
-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name
=asyncio_0
2026-01-26T00:36:38.942407Z [info     ] 🛠 检测到验证报告，进入优化模式...           [__src__agent__graph_08] api_variant=local_dev assista
n                                                                                                                                          n
```

```
2026-01-26T00:40:24.467914Z [info     ] ✅优化成功! 生成优化类代码                [__src__agent__graph_08] api_variant=local_dev assistant__
id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8
-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread
_name=asyncio_0
2026-01-26T00:40:24.468913Z [info     ] 🔬 开始代码验证...                    [__src__agent__graph_08] api_variant=local_dev assistant_id=fe
096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e
-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name
=asyncio_0
```

```
2026-01-26T00:41:41.339266Z [info     ] ✅验证完成!                          [__src__agent__graph_08] api_variant=local_dev assistant_id=fe099
6781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74dd8-1a2e-4
be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=a
syncio_0
2026-01-26T00:41:41.340255Z [info     ] 📊 分析验证报告，决定是否需要继续优化...         [__src__agent__graph_08] api_variant=local_dev assi
stant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4
a74dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17
thread_name=asyncio_0
2026-01-26T00:41:41.340255Z [info     ] ✅代码质量良好，无需进一步优化                [__src__agent__graph_08] api_variant=local_dev assistaa
nt_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=optimize_and_evaluate request_id=a4a74
dd8-1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thr
ead_name=asyncio_0
2026-01-26T00:41:41.348761Z [info     ] 📄 生成最终技能原语库...                [__src__agent__graph_08] api_variant=local_dev assistant_id
=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=generate_output request_id=a4a74dd8-1a2e-4b
e9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_name=as
yncio_0
2026-01-26T00:41:41.349761Z [info     ] 🎉 生成成功! 输出文件: turtlebot3_example_01.py [__src__agent__graph_08] api_variant=local_dev assis
tant_id=fe096781-5601-53d2-b2f6-0d3403f7e9ca graph_id=agent langgraph_api_version=0.4.31 langgraph_node=generate_output request_id=a4a74dd8-
1a2e-4be9-bc92-994e3b081b49 run_attempt=1 run_id=019bf79f-14e0-709d-be8c-4fba2af19ebf thread_id=0f801d10-e3c3-4301-8d86-67d38c510a17 thread_
name=asyncio_0
```